

## MCCLELLAN-PALOMAR, CA, USA

WMO: 722927

Lat: 33.128N

Lon: 117.279W

Elev: 100

StdP: 100.13

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 03177

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 12            | 6.1        | 7.2 | -11.2                         | 1.5 | 15.3 | -8.2 | 1.9 | 14.4 | 7.6                   | 16.1 | 6.6 | 15.1 | 1.4                   | 80   | 0.337 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.0                    | 29.1            | 17.2 | 27.3 | 17.8 | 25.8 | 17.9 | 21.7                | 25.3 | 20.9 | 24.3 | 20.4 | 23.6 | 2.9                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.3                            | 15.2 | 23.5 | 19.6 | 14.5 | 23.1 | 19.0 | 14.0 | 22.6 | 63.4          | 25.4 | 61.0 | 24.4 | 59.0 | 23.8 | 24.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |  |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|--|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |  |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |  |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |  |
| 6.0               | 5.3  | 4.8 | DB                         | 3.6 | 34.6               | 0.9 | 2.8                                                | 2.9  | 36.7       | 2.4  | 38.3       | 1.9  | 39.9       | 1.2  | 41.9 |  |
|                   |      |     | WB                         | 0.1 | 22.8               | 1.1 | 1.0                                                | -0.7 | 23.5       | -1.4 | 24.0       | -2.0 | 24.6       | -2.8 | 25.3 |  |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 16.9   | 13.8  | 13.5  | 14.3  | 15.3  | 16.8  | 18.3  | 20.7  | 21.3  | 20.7  | 18.8  | 16.0  | 13.4 |
|                                                                   | DBStd       | 3.68   | 2.87  | 2.59  | 2.35  | 2.40  | 1.95  | 1.62  | 1.74  | 1.68  | 2.35  | 2.62  | 2.72  | 2.39 |
|                                                                   | HDD10.0     | 4      | 2     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1    |
|                                                                   | HDD18.3     | 846    | 145   | 139   | 129   | 97    | 56    | 18    | 1     | 0     | 2     | 23    | 81    | 155  |
|                                                                   | CDD10.0     | 2531   | 119   | 98    | 133   | 158   | 210   | 249   | 332   | 349   | 322   | 273   | 181   | 106  |
|                                                                   | CDD18.3     | 330    | 3     | 3     | 3     | 6     | 8     | 18    | 74    | 91    | 74    | 38    | 11    | 1    |
|                                                                   | CDH23.3     | 1097   | 26    | 23    | 31    | 46    | 53    | 27    | 118   | 187   | 271   | 217   | 86    | 12   |
| Wind                                                              | WSAvg       | 1.9    | 1.7   | 2.0   | 2.1   | 2.3   | 2.4   | 2.2   | 2.0   | 1.9   | 1.8   | 1.7   | 1.6   | 1.7  |
|                                                                   | PrecAvg     | 287    | 58    | 59    | 40    | 22    | 7     | 3     | 3     | 1     | 4     | 14    | 22    | 42   |
|                                                                   | PrecMax     | 584    | 252   | 255   | 160   | 53    | 29    | 20    | 43    | 9     | 28    | 126   | 76    | 203  |
|                                                                   | PrecMin     | 90     | 4     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Precipitation                                                     | PrecStd     | 136    | 62    | 56    | 39    | 16    | 8     | 5     | 9     | 2     | 6     | 27    | 21    | 42   |
|                                                                   | DB          | 26.2   | 26.7  | 27.0  | 28.1  | 29.0  | 26.4  | 27.9  | 29.0  | 32.0  | 31.3  | 29.1  | 24.8  |      |
|                                                                   | MCWB        | 13.2   | 13.1  | 13.6  | 13.8  | 14.3  | 19.5  | 21.0  | 20.3  | 21.3  | 16.8  | 14.3  | 11.9  |      |
|                                                                   | DB          | 23.2   | 22.6  | 23.0  | 23.8  | 23.2  | 23.2  | 26.0  | 26.9  | 28.2  | 28.2  | 25.8  | 21.5  |      |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | MCWB        | 12.0   | 11.8  | 12.8  | 13.6  | 15.7  | 18.6  | 20.4  | 20.3  | 19.8  | 15.3  | 13.4  | 11.3  |      |
|                                                                   | DB          | 21.1   | 20.0  | 20.1  | 21.1  | 20.9  | 22.0  | 24.5  | 25.3  | 26.3  | 25.8  | 22.8  | 19.3  |      |
|                                                                   | MCWB        | 11.5   | 11.8  | 12.7  | 13.6  | 15.9  | 17.9  | 20.1  | 20.2  | 20.0  | 15.5  | 12.6  | 11.4  |      |
|                                                                   | DB          | 18.7   | 17.7  | 17.9  | 18.9  | 19.1  | 21.0  | 23.4  | 24.0  | 24.3  | 23.0  | 20.5  | 17.6  |      |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | MCWB        | 11.2   | 11.8  | 13.0  | 13.9  | 15.1  | 17.3  | 19.6  | 19.8  | 19.4  | 16.5  | 13.0  | 11.6  |      |
|                                                                   | WB          | 15.6   | 15.8  | 16.5  | 16.8  | 18.1  | 20.0  | 22.3  | 22.3  | 22.7  | 21.0  | 17.3  | 16.2  |      |
|                                                                   | MCDB        | 19.2   | 18.5  | 20.9  | 21.4  | 22.1  | 24.3  | 25.7  | 26.0  | 28.3  | 24.3  | 20.2  | 19.1  |      |
|                                                                   | WB          | 14.6   | 14.6  | 15.3  | 15.7  | 17.0  | 19.0  | 21.2  | 21.4  | 21.7  | 19.5  | 16.3  | 15.1  |      |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | MCDB        | 17.7   | 17.6  | 18.9  | 19.5  | 20.4  | 22.6  | 24.4  | 24.8  | 25.8  | 23.5  | 19.9  | 17.2  |      |
|                                                                   | WB          | 13.9   | 13.9  | 14.6  | 15.1  | 16.4  | 18.3  | 20.5  | 20.7  | 20.9  | 18.5  | 15.6  | 14.3  |      |
|                                                                   | MCDB        | 16.8   | 16.9  | 17.8  | 18.5  | 19.5  | 21.3  | 23.7  | 24.0  | 24.4  | 22.1  | 19.1  | 16.6  |      |
|                                                                   | WB          | 13.2   | 13.2  | 14.0  | 14.5  | 15.7  | 17.6  | 20.0  | 20.2  | 20.2  | 17.7  | 15.0  | 13.4  |      |
| Mean Daily Temperature Range                                      | MCDB        | 16.3   | 16.4  | 16.8  | 17.7  | 18.5  | 20.4  | 22.8  | 23.4  | 23.6  | 20.9  | 18.3  | 15.9  |      |
|                                                                   | MDBR        | 8.8    | 7.9   | 6.9   | 6.4   | 5.0   | 4.5   | 4.7   | 5.0   | 5.9   | 7.2   | 8.5   | 8.7   |      |
|                                                                   | 5% DB       | 12.8   | 12.1  | 11.1  | 11.2  | 8.2   | 6.3   | 6.2   | 6.7   | 8.7   | 12.2  | 12.5  | 11.8  |      |
|                                                                   | MCWBR       | 6.8    | 6.8   | 6.2   | 5.9   | 4.3   | 2.9   | 2.6   | 3.0   | 4.2   | 7.0   | 7.3   | 7.2   |      |
| Clear-Sky Solar Irradiance                                        | 5% WB       | 8.6    | 8.4   | 7.6   | 7.6   | 6.2   | 6.1   | 5.8   | 6.0   | 7.0   | 8.2   | 8.7   | 8.1   |      |
|                                                                   | MCWBR       | 5.5    | 5.4   | 4.6   | 4.3   | 3.4   | 2.9   | 2.5   | 2.8   | 3.4   | 5.3   | 5.8   | 5.7   |      |
|                                                                   | taub        | 0.307  | 0.317 | 0.334 | 0.346 | 0.365 | 0.362 | 0.391 | 0.386 | 0.357 | 0.355 | 0.315 | 0.317 |      |
|                                                                   | taud        | 2.569  | 2.557 | 2.493 | 2.463 | 2.435 | 2.449 | 2.444 | 2.435 | 2.500 | 2.479 | 2.581 | 2.574 |      |
| All-Sky Solar Radiation                                           | Ebn at Noon | 917    | 941   | 946   | 941   | 921   | 918   | 890   | 891   | 908   | 883   | 901   | 885   |      |
|                                                                   | Edh at Noon | 80     | 90    | 103   | 111   | 115   | 113   | 114   | 113   | 101   | 96    | 79    | 76    |      |
|                                                                   | RadAvg      | 3.27   | 4.14  | 5.47  | 6.60  | 6.99  | 7.36  | 7.25  | 6.95  | 5.95  | 4.58  | 3.58  | 2.96  |      |
| Historical Trends                                                 | RadStd      | 0.21   | 0.38  | 0.29  | 0.35  | 0.49  | 0.46  | 0.42  | 0.23  | 0.27  | 0.32  | 0.22  | 0.21  |      |

## Historical Trends

|                         |       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                     | (e)   | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| Station Only            | +0.93 | N/A   | N/A     | +1.36  | +0.95   | +0.88 | N/A   | -210        | +306    | +132    |         |
| Regional (45 neighbors) | +0.60 | N/A   | -1.62   | +0.66  | N/A     | N/A   | N/A   | -98         | +211    | +110    |         |

Nomenclature: See separate page